

Christopher Musco

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Education

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. Candidate, Computer Science – Theory of Computation

2013 – present

Advisors: Jonathan Kelner, Martin Rinard

S.M. Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

Yale University

New Haven, CT

B.S. Applied Mathematics, B.S. Computer Science

2008 – 2012

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (under Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (under Daniel Spielman)

Other Academic Positions

Yahoo Labs

New York, NY

Research Intern, Scalable Machine Learning Group

Summer 2015

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on theoretical matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

Research Areas

theory of algorithms ◦ machine learning ◦ randomized numerical linear algebra ◦ low rank approximation
dimensionality reduction ◦ sketching and streaming ◦ algorithmic graph theory ◦ iterative matrix algorithms

Publications

Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.

Conference Papers

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *ICML* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *NIPS* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers).

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. Invited to SICOMP Special Issue.

Other

Provably Useful Kernel Matrix Approximation in Linear Time. Cameron Musco, Christopher Musco. *In submission.*

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *In submission.*

Ridge Leverage Scores for Low-Rank Approximation. Michael B. Cohen, Cameron Musco, Christopher Musco. *In submission.*

Teaching and Mentorship

Massachusetts Institute of Technology

Teaching Assistant, Advanced Algorithms

2016

Hold weekly office hours, write problem sets, and compile notes and resources for an updated version of this capstone graduate algorithms course at MIT.

Workshop Leader, Technical Communication Skills for Graduate Students

2016

Lead weekly workshops to supplement lecture material with hands-on writing and presentation exercises.

Communication Advisor

2015 – present

Helping to establish an EECS Communication Lab to provided over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively.

Research Science Institute at MIT

Summer Research Advisor

2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Chris' project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Yale University

Teaching Assistant, Introduction to Computer Science

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Teaching Assistant, Introduction to Programming

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

Grader, Introductory Calculus

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Talks

Principal Component Projection without Principal Component Analysis

International Conference on Machine Learning (ICML)

June 2016

Ridge Leverage Score Sampling

University of Utah, Data Group Meeting

Jan. 2016

Randomized Block Krylov Methods

Neural Information Processing Systems (NIPS) Poster Session

Dec. 2015

Large Scale Column Subset Selection

Yahoo Labs, Science Week Poster Session

July 2015

Dimensionality Reduction for K-Means Clustering

IBM T.J. Watson, Mathematical Sciences Research Seminar

Aug. 2015

Symposium on Theory of Computing (STOC)

June 2015

MIT, Theory Lunch

June 2014

Principled Sampling for Anomaly Detection

MIT, Defense Advanced Research Projects Agency Site Visit

Mar. 2015

Network and Distributed System Security Symposium (NDSS)

Feb. 2015

Uniform Sampling for Matrix Approximation

Yahoo Labs, Scalable Machine Learning Research Seminar

June 2015

Single Pass Spectral Sparsification in Dynamic Streams

MIT, Annual Sublinear Algorithms Day Poster Session

Apr. 2015

Harvard University, Theory Seminar

Nov. 2014

Foundations of Computer Science (FOCS)

Oct. 2014

Service

External Reviewer.....

Conference on Learning Theory (COLT) 2016

Symposium on Discrete Algorithms (SODA) 2016

IEEE Transactions on Signal Processing 2016

Neural Information Processing Systems (NIPS) 2015

European Symposia on Algorithms (ESA) 2015

Foundations of Computer Science (FOCS) 2015

International Colloquium on Automata, Languages and Programming (ICALP) 2015

Network and Distributed System Security Symposium (NDSS) 2015

Honors and Awards

Japan National Institute of Informatics 2015

Invited participant at "Recent Advances in Randomized Numerical Linear Algebra" meeting.

National Science Foundation 2014

Graduate Research Fellowship.

Yale University 2012

Cum Laude, Distinction in Both Majors.

Professional Experience

Redfin **Seattle, WA**

Software Engineer, Data Team

2012 – 2014

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.

Elysium Digital **Cambridge, MA**

Summer Consultant

Summer 2011

Provided technical expertise in software patent cases at leading litigation consulting company.

Amicus **New Haven, CT**

Software Developer

2010 – 2011

Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.

Moody's Investors Service **New York, NY**

Intern, Asset Finance Group

Summer 2009

Established and monitored bond ratings on securitizations in three asset classes. Wrote code to automate analysis for student loan data and to compile records on aircraft leases.

U.S. Department of Agriculture **East Greenwich, RI**

Intern, Natural Resources Conservation Service

Summer 2007

Collected and analyzed data for two major state conservation and research projects. Integrated mapping software and database tools to develop a system for compiling and presenting soil data.